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Topic 11 – FDMs for Black-Scholes equations

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- 2 FDMs for Black-Scholes PDE
 - Approximating derivatives
 - Backward Euler scheme
 - Forward Euler scheme
 - Crank-Nicolson (CN) Method
 - Risk Management

Finite difference methods (FDMs)

- FDMs are widely used for numerically solving various types of PDEs.
- In FDMs, the solution is approximated at discrete grid points in space and/or time.
- The constructions of FDMs rely on truncated Taylor series expansions. Derivatives are approximated using differences between function values.
- FDMs are broadly classified into two categories: **explicit** & **implicit** methods.
- For time-dependent PDEs, **implicit** FDMs typically require solving a (possibly large) linear system at each time step.
- **Explicit** FDMs are computationally more attractive, as they do not involve solving linear systems.
- A key drawback of explicit methods is that they often require strict stability conditions. Implicit methods are generally more stable.
- The accuracy depends on the order of the method and the nature of the PDE being solved.

The Black–Scholes PDE

Consider the Black–Scholes PDE: for $0 < t < T$ and $0 < S < \infty$,

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV.$$

Examples of the boundary and terminal (at the final-time T) conditions.

- For a call option $V = c$,

$$c(0, t) = 0 \quad \text{for } 0 < t < T,$$

$$c(S, t) \approx S - Ke^{-r(T-t)} \quad \text{as } S \rightarrow \infty \text{ for } 0 < t < T,$$

$$c(S, T) = \max\{0, S - K\} \quad \text{for } 0 < S < \infty.$$

- For a put option $V = p$,

$$p(0, t) = Ke^{-r(T-t)} \quad \text{for } 0 < t < T,$$

$$p(S, t) \approx 0 \quad \text{as } S \rightarrow \infty \text{ for } 0 < t < T,$$

$$p(S, T) = \max\{0, K - S\} \quad \text{for } 0 < S < \infty.$$

In general, solving the Black–Scholes PDE exactly is difficult, so we use explicit and implicit FDMs to obtain numerical (approximate) solutions.

Preliminary steps

Step 1. Truncate the semi-infinite domain $[0, \infty)$ to a finite interval $[0, S_{\max}]$. $S_{\max}$ is chosen sufficiently large such that $S_{\max} > K$.

Step 2. Define the grid points. Uniformly divide the intervals $[0, S_{\max}]$ and $[0, T]$ into $N_S + 1$ and N_t subintervals, respectively. Then define the grid points as

$$(S_j, t_n) = (j \Delta S, n \Delta t), \quad \text{where } \Delta S = \frac{S_{\max}}{N_S + 1}, \quad \Delta t = \frac{T}{N_t},$$

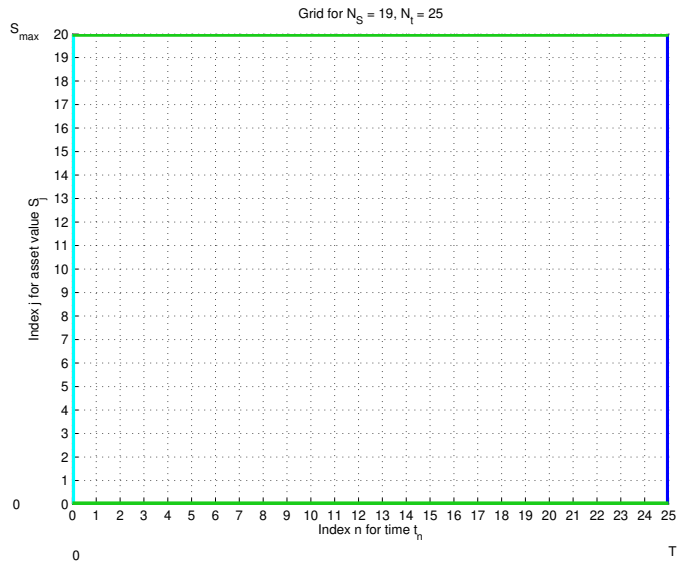
for $j = 0, 1, 2, \dots, N_S + 1$ and $n = 0, 1, 2, \dots, N_t$.

Step 3. At each grid point (S_j, t_n) , the Black–Scholes PDE becomes:

$$\frac{\partial V}{\partial t}(S_j, t_n) + r S_j \frac{\partial V}{\partial S}(S_j, t_n) + \frac{1}{2} \sigma^2 S_j^2 \frac{\partial^2 V}{\partial S^2}(S_j, t_n) = r V(S_j, t_n).$$

Key Question: How can we approximate the partial derivatives?

Example of a grid points



V at the grid points

From the boundary and terminal conditions (say for the case of a call option where $c(0, t) = 0$ and $c(S_{\max}, t) = S_{\max} - Ke^{-r(T-t)}$ for $0 < t < T$, and $c(S, T) = \max\{0, S - K\}$ for $0 < S < S_{\max}$).

- V is known at the grid points on the right-vertical boundary
- V is known at the grid points on the lower and upper horizontal boundaries
- V is unknown at the interior grid points, as well as those on the left vertical segment. That is, $V(S_j, t_n)$ is unknown for $j = 1, 2, \dots, N_S$ and $n = 0, 1, \dots, N_t - 1$.
- We approximate the unknown $V(S_j, t_n)$ by a finite difference solution V_j^n .
- We define V_j^n via three different numerical approaches:
 - Backward Euler FDM (Explicit),
 - Forward Euler FDM (Implicit),
 - Crank-Nicolson FDM (Implicit).

Approximating derivatives in the t direction (Topic 07)

Given a function f , δ is a non-zero constant ($|\delta|$ is usually small). From the Taylor series expansion,

$$\bullet f(q + \delta) = f(q) + \delta f'(q) + O(\delta^2) \implies f'(q) = \frac{f(q+\delta) - f(q)}{\delta} + O(\delta).$$

Therefore, $f'(q)$ can be approximated as:

$$f'(q) \approx \frac{f(q + \Delta t) - f(q)}{\Delta t} \quad \text{with } O(\Delta t) \text{ error}$$

(The difference quotient is taken in the *forward* direction, **first-order forward approximation.**) Or

$$f'(q) \approx \frac{f(q) - f(q - \Delta t)}{\Delta t} \quad \text{with } O(\Delta t) \text{ error}$$

(The difference quotient is taken in the *backward* direction, **first-order backward approximation.**)

Approximating derivatives in the S direction (Topic 07)

Again, from the Taylor series expansion,

- $f(q + \delta) = f(q) + \delta f'(q) + \frac{\delta^2}{2} f''(q) + \frac{\delta^3}{6} f'''(q) + O(\delta^4)$, and so,
 - $f(q + \Delta S) = f(q) + \Delta S f'(q) + \frac{(\Delta S)^2}{2} f''(q) + \frac{\Delta S^3}{6} f'''(q) + O((\Delta S)^4)$,
 - $f(q - \Delta S) = f(q) - \Delta S f'(q) + \frac{(\Delta S)^2}{2} f''(q) - \frac{\Delta S^3}{6} f'''(q) + O((\Delta S)^4)$,

Thus,

- $f''(q) = \frac{f(q+\Delta S) - 2f(q) + f(q-\Delta S)}{(\Delta S)^2} + O((\Delta S)^2)$,
- $f'(q) = \frac{f(q+\Delta S) - f(q-\Delta S)}{2\Delta S} + O((\Delta S)^2)$.

Therefore,

$$f''(q) \approx \frac{f(q + \Delta S) - 2f(q) + f(q - \Delta S)}{(\Delta S)^2} \quad \text{with } O((\Delta S)^2) \text{ error,}$$

(second central difference approximation) and

$$f'(q) \approx \frac{f(q + \Delta S) - f(q - \Delta S)}{2\Delta S} \quad \text{with } O((\Delta S)^2) \text{ error,}$$

(central difference approximation).

Backward Euler scheme

Aim: Approximate $V(S_j, t_n)$ by V_j^n using a backward Euler FDM.

Recall that the Black–Scholes PDE at the grid point (S_j, t_n) is:

$$\frac{\partial V}{\partial t}(S_j, t_n) + rS_j \frac{\partial V}{\partial S}(S_j, t_n) + \frac{1}{2}\sigma^2 S_j^2 \frac{\partial^2 V}{\partial S^2}(S_j, t_n) = rV(S_j, t_n).$$

Approximate the partial derivatives as:

$$\frac{\partial V}{\partial t}(S_j, t_n) \approx \frac{V(S_j, t_n) - V(S_j, t_{n-1})}{\Delta t} \quad \text{backward with } O(\Delta t) \text{ error,}$$

$$\frac{\partial V}{\partial S}(S_j, t_n) \approx \frac{V(S_{j+1}, t_n) - V(S_{j-1}, t_n)}{2\Delta S} \quad \text{central difference with } O((\Delta S)^2) \text{ error,}$$

$$\frac{\partial^2 V}{\partial S^2}(S_j, t_n) \approx \frac{V(S_{j+1}, t_n) - 2V(S_j, t_n) + V(S_{j-1}, t_n)}{(\Delta S)^2} \quad \text{with } O((\Delta S)^2) \text{ error.}$$

The last one is the second central difference.

Backward Euler scheme (continued)

Therefore,
$$\frac{V(S_j, t_n) - V(S_j, t_{n-1})}{\Delta t} + rS_j \frac{V(S_{j+1}, t_n) - V(S_{j-1}, t_n)}{2\Delta S} + \frac{1}{2}\sigma^2 S_j^2 \frac{V(S_{j+1}, t_n) - 2V(S_j, t_n) + V(S_{j-1}, t_n)}{(\Delta S)^2} \approx rV(S_j, t_n).$$

The backward Euler finite difference solution $V_j^n \approx V(S_j, t_n)$ is then defined as:

$$\frac{V_j^n - V_j^{n-1}}{\Delta t} + rS_j \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S} + \frac{1}{2}\sigma^2 S_j^2 \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{(\Delta S)^2} = rV_j^n,$$

for $j = 1, 2, 3, \dots, N_S$ and $n = N_t, N_t - 1, \dots, 2, 1$.

Boundary conditions: $V_0^n = V(0, t_n)$, $V_{N_S+1}^n = V(S_{\max}, t_n)$, $n = 0, 1, \dots, N_t - 1$.

Terminal condition: $V_j^{N_t} = V(S_j, T)$, $j = 0, 1, \dots, N_S + 1$.

- In the *stable* case, the error satisfies:

$\max_{j,n} |V_j^n - V(S_j, t_n)| \leq C(\Delta t + (\Delta S)^2)$, which implies that the backward Euler scheme is $O(\Delta t + (\Delta S)^2)$ accurate.

- Is this scheme always stable? **Not necessarily.** Stability conditions will be discussed later.

Implementations

Multiplying through by Δt ,

$$V_j^n - V_j^{n-1} + \frac{rS_j\Delta t}{2\Delta S}(V_{j+1}^n - V_{j-1}^n) + \frac{\sigma^2 S_j^2 \Delta t}{2(\Delta S)^2}(V_{j+1}^n - 2V_j^n + V_{j-1}^n) = r\Delta t V_j^n.$$

Setting $\rho = \frac{\Delta t}{2(\Delta S)^2}$ and rearranging the terms,

$$(1 - r\Delta t - 2\sigma^2 S_j^2 \rho)V_j^n - V_j^{n-1} + (\sigma^2 S_j^2 \rho + rS_j \Delta S \rho)V_{j+1}^n + (\sigma^2 S_j^2 \rho - rS_j \Delta S \rho)V_{j-1}^n = 0,$$

for $j = 1, 2, 3, \dots, N_S$ and $n = N_t, N_t - 1, \dots, 2, 1$. For convenience, let

$$\alpha_j = (\sigma^2 S_j^2 - rS_j \Delta S)\rho, \quad \gamma_j = (\sigma^2 S_j^2 + rS_j \Delta S)\rho, \\ \beta_j = 1 - r\Delta t - 2\sigma^2 S_j^2 \rho = 1 - 2(r(\Delta S)^2 + \sigma^2 S_j^2)\rho.$$

Then,

$$V_j^{n-1} = \alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n, \quad 1 \leq n \leq N_t.$$

- This is an explicit formula for unknowns in terms of known, and so, the proposed backward Euler scheme is **explicit**.

Implementations (continued)

As an example, in the case of a call option, where

$$V(0, t) = 0 \quad \text{and} \quad V(S_{\max}, t) \approx S_{\max} - Ke^{-r(T-t)} \quad \text{for } 0 < t < T,$$

$$V(S, T) = \max\{S - K, 0\} \quad \text{for } 0 < S < S_{\max},$$

we have

$$V_0^n = 0 \quad \text{and} \quad V_{N_S+1}^n = S_{\max} - Ke^{-r(T-t_n)} \quad \text{for } n = 0, 1, \dots, N_t - 1,$$

$$V_j^{N_t} = \max\{S_j - K, 0\} \quad \text{for } j = 0, 1, \dots, N_S + 1.$$

So,

$$V_j^{N_t-1} = \alpha_j \max\{S_{j-1} - K, 0\} + \beta_j \max\{S_j - K, 0\} + \gamma_j \max\{S_{j+1} - K, 0\},$$

for $j = 1, 2, \dots, N_S$. Thus, $V_j^{N_t-1}$ is now known for all $j = 0, \dots, N_S + 1$.

We then proceed recursively to compute V_j^{n-1} for $n = N_t - 1, \dots, 1$.

- The algorithm can be implemented in MATLAB (`bspdeexp.m`) or Python (`bspdeexp` in `bspde.py`)

Stability

From the backward Euler scheme,

$$V_j^{n-1} = \alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n,$$

for $j = 1, 2, 3, \dots, N_S$ and $n = N_t, N_t - 1, \dots, 2, 1$. Let

$$G = \max_{j,n} \{|V_j^{N_t}|, |V_0^n|, |V_{N_S+1}^n|\} = \max_{j,n} \{|V(S_j, T)|, |V(0, t_n)|, |V(S_{\max}, t_n)|\}.$$

G is known from the boundary and terminal (final-time) conditions.

- The proposed explicit scheme is **stable** if $\max_j |V_j^{n-1}| \leq C G$ for $n = N_t, \dots, 2, 1$, where C is a constant independent of Δt and ΔS . For example, if $|\alpha_j| + |\beta_j| + |\gamma_j| \leq 1$ for all $j = 1, 2, \dots, N_S$, then $|V_j^{N_t-1}| \leq (|\alpha_j| + |\beta_j| + |\gamma_j|) \max_j |V_j^{N_t}| \leq G \implies \max_j |V_j^{N_t-1}| \leq G$, and by recursion, $\max_j |V_j^{n-1}| \leq G$ for $1 \leq n \leq N_t - 1$.

Next, we discuss the constraints required to ensure that $|\alpha_j| + |\beta_j| + |\gamma_j| \leq 1$.

Stability (continued)

Recall that,

$$\alpha_j = (\sigma^2 S_j^2 - r S_j \Delta S) \rho, \quad \gamma_j = (\sigma^2 S_j^2 + r S_j \Delta S) \rho, \quad \beta_j = 1 - 2(r(\Delta S)^2 + \sigma^2 S_j^2) \rho$$

We impose the following assumptions on ρ and σ :

$$\sigma^2 \geq r \quad \& \quad 2\rho = \frac{\Delta t}{(\Delta S)^2} \leq \frac{1}{r(\Delta S)^2 + \sigma^2 S_j^2} \quad \text{for } 1 \leq j \leq N_s \quad (*)$$

Then, $\alpha_j \geq r S_j (S_j - \Delta S) \rho \geq 0$ and $\beta_j = 1 - 2(r(\Delta S)^2 + \sigma^2 S_j^2) \rho \geq 0$.

Therefore,

$$\bullet \quad |\alpha_j| + |\beta_j| + |\gamma_j| = \alpha_j + \beta_j + \gamma_j = 1 - 2r(\Delta S)^2 \rho = 1 - 2r\Delta t \leq 1.$$

The mesh constraint in (*) is somehow restrictive. The backward Euler scheme in this case is $O((\Delta S)^2)$ accurate because

$$\max_{j,n} |V_j^n - V(x_j, t_n)| = O(\Delta t + (\Delta S)^2) = O((\Delta S)^2).$$

The inequality $|\alpha_j| + |\beta_j| + |\gamma_j| \leq 1$ is a sufficient condition for stability; however, it may be relaxed in certain cases.

Examples (Explicit FDM `bspdetst.m`)

Consider a call option with

$$r = 0.05, \quad \sigma = 0.3, \quad T = 2, \quad S_{\max} = 8, \quad K = 2.$$

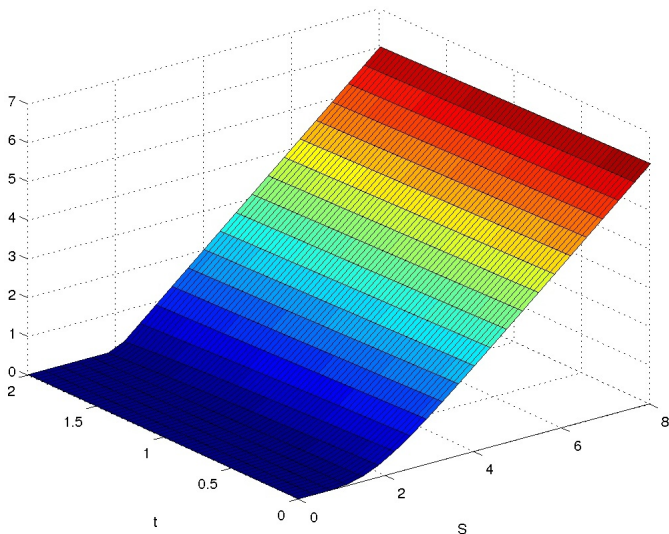
Use a space discretization with $N_S = 21$, so

$$\Delta S = \frac{S_{\max}}{N_S + 1} = \frac{8}{22} \approx 0.3636$$

- ① Number of time steps $N_t = 60 \implies \Delta t = \frac{T}{N_t} = \frac{2}{60} \approx 0.0333$
 - The numerical solution is stable.
- ② Number of time steps $N_t = 55 \implies \Delta t = \frac{T}{N_t} = \frac{2}{55} \approx 0.0364$
 - The numerical solution becomes unstable (oscillations).
- ③ Number of time steps $N_t = 53 \implies \Delta t = \frac{T}{N_t} = \frac{2}{53} \approx 0.0364$
 - The numerical solution becomes more unstable (more oscillations).

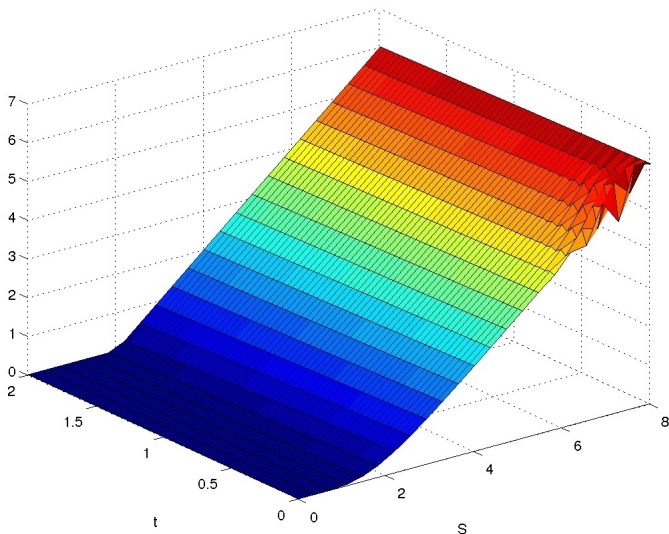
Graphical plot of the approximate solution (1)

Explicit Finite Difference Solution, $N_S = 21$, $N_t = 60$

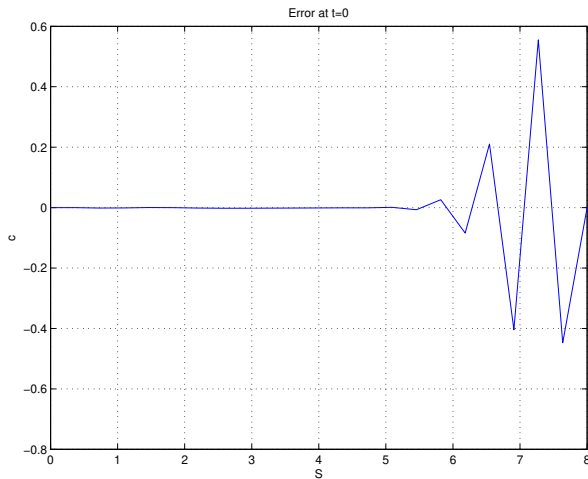


Graphical plot of the approximate solution (2)

Explicit Finite Difference Solution, $N_S = 21$, $N_t = 55$

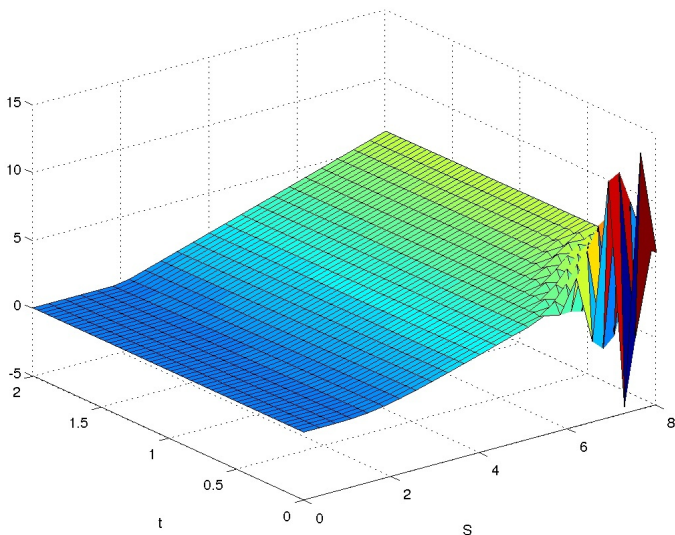


Error profile at $t = 0$ (2)



Graphical plot of the approximate solution (3)

Explicit Finite Difference Solution, $N_S = 21$, $N_t = 53$



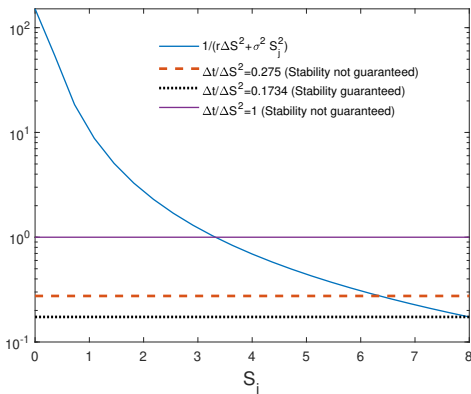
Numerical instability in example (2)

Let us justify the instability in Example (2).

- $\sigma = 0.3$ and $r = 0.05 \implies \sigma^2 > r$ (as required)
- $N_t = 55$ and $\Delta S = 8/22 \implies \frac{\Delta t}{(\Delta S)^2} \approx 0.2750$
- We require $\frac{\Delta t}{(\Delta S)^2} = \min_{j=1,\dots,N_s} \frac{1}{r(\Delta S)^2 + \sigma^2 S_j^2} \approx 0.1734$.

$\frac{\Delta t}{(\Delta S)^2} \approx 0.2750 > 0.1734$, so the stability is not guaranteed.

The constraint $\frac{\Delta t}{(\Delta S)^2} \leq \min_{j=1,\dots,N_s} \frac{1}{r(\Delta S)^2 + \sigma^2 S_j^2}$ is a sufficient, though not necessary, condition for stability. It may be relaxed slightly, but not significantly.



Forward Euler scheme

We use a *forward* difference instead of a *backward* difference to approximate $\partial V / \partial t$. Recall that, the backward Euler scheme is:

$$\frac{V_j^n - V_j^{n-1}}{\Delta t} + rS_j \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S} + \frac{1}{2}\sigma^2 S_j^2 \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{(\Delta S)^2} = rV_j^n,$$

for $j = 1, 2, 3, \dots, N_S$ and $n = N_t, N_t - 1, \dots, 2, 1$.

The forward Euler scheme is defined as:

$$\frac{V_j^{n+1} - V_j^n}{\Delta t} + rS_j \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S} + \frac{1}{2}\sigma^2 S_j^2 \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{(\Delta S)^2} = rV_j^n,$$

for $j = 1, 2, 3, \dots, N_S$ and $n = N_t - 1, N_t - 2, \dots, 2, 1, 0$.

As in the backward Euler case, multiply through by Δt ,

$$V_j^{n+1} - V_j^n + rS_j \frac{\Delta t}{2\Delta S} (V_{j+1}^n - V_{j-1}^n) + \frac{1}{2}\sigma^2 S_j^2 \frac{\Delta t}{(\Delta S)^2} (V_{j+1}^n - 2V_j^n + V_{j-1}^n) = r\Delta t V_j^n,$$

Forward Euler scheme (continued)

Rearranging the terms,

$$V_j^{n+1} - \alpha_j V_{j-1}^n - \beta_j V_j^n - \gamma_j V_{j+1}^n = 0,$$

where

$$\alpha_j = \frac{r S_j \Delta t}{2 \Delta S} - \frac{\sigma^2 S_j^2 \Delta t}{2 (\Delta S)^2} = j(r - \sigma^2 j) \frac{\Delta t}{2}$$

$$\beta_j = 1 + r \Delta t + \frac{\sigma^2 S_j^2 \Delta t}{(\Delta S)^2} > 0$$

$$\gamma_j = -\frac{\sigma^2 S_j^2 \Delta t}{2 (\Delta S)^2} - \frac{r S_j \Delta t}{2 \Delta S} < 0$$

Therefore, for $j = 1, 2, \dots, N_S$ and $n = N_t - 1, N_t - 2, \dots, 1, 0$, we have

$$\alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n = V_j^{n+1}.$$

For later use: if $\sigma^2 \geq r \implies |\alpha_j| + |\gamma_j| = -\alpha_j - \gamma_j = \frac{\sigma^2 S_j^2 \Delta t}{(\Delta S)^2} < \beta_j$.

Matrix form, $\alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n = V_j^{n+1},$

$$\alpha_1 V_0^n + \beta_1 V_1^n + \gamma_1 V_2^n = V_1^{n+1}$$

$$\alpha_2 V_1^n + \beta_2 V_2^n + \gamma_2 V_3^n = V_2^{n+1}$$

$$\alpha_3 V_2^n + \beta_3 V_3^n + \gamma_3 V_4^n = V_3^{n+1}$$

$$\vdots$$

$$\Rightarrow$$

$$\alpha_{N_S-1} V_{N_S-2}^n + \beta_{N_S-1} V_{N_S-1}^n + \gamma_{N_S-1} V_{N_S}^n = V_{N_S-1}^{n+1}$$

$$\alpha_{N_S} V_{N_S-1}^n + \beta_{N_S} V_{N_S}^n + \gamma_{N_S} V_{N_S+1}^n = V_{N_S}^{n+1}$$

$$\underbrace{\begin{bmatrix} \beta_1 & \gamma_1 & & & \\ \alpha_2 & \beta_2 & \gamma_2 & & \\ & \alpha_3 & \beta_3 & \gamma_3 & \\ & & \ddots & \ddots & \ddots \\ & & & \alpha_{N_S-1} & \beta_{N_S-1} & \gamma_{N_S-1} \\ & & & & \alpha_{N_S} & \beta_{N_S} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} V_1^n \\ V_2^n \\ V_3^n \\ \vdots \\ V_{N_S-1}^n \\ V_{N_S}^n \end{bmatrix}}_{\mathbf{V}^n} = \underbrace{\begin{bmatrix} V_1^{n+1} \\ V_2^{n+1} \\ V_3^{n+1} \\ \vdots \\ V_{N_S-1}^{n+1} \\ V_{N_S}^{n+1} \end{bmatrix}}_{\mathbf{b}^n} - \underbrace{\begin{bmatrix} \alpha_1 V_0^n \\ 0 \\ \vdots \\ \vdots \\ 0 \\ \gamma_{N_S} V_{N_S+1}^n \end{bmatrix}}_{\mathbf{b}^n}$$

Matrix form (continued)

Therefore, we have to solve the linear system $\mathbf{A}\mathbf{V}^n = \mathbf{b}^n$ for $n = N_t - 1, N_t - 2, \dots, 1, 0$, see [bspdeimp.m](#) or [bspde.py](#).

The coefficient matrix $\mathbf{A}$ is

- tridiagonal, not symmetric (because $\alpha_{j+1} \neq \gamma_j$), not Toeplitz,
- independent of n
- Strictly diagonally dominant if $\sigma^2 \geq r$ (because $\beta_j > |\alpha_j| + |\gamma_j|$ for $j = 1, \dots, N_S$), and hence, $\mathbf{A}$ is non-singular. Consequently, the linear system $\mathbf{A}\mathbf{V}^n = \mathbf{b}^n$ has a unique solution for each n .

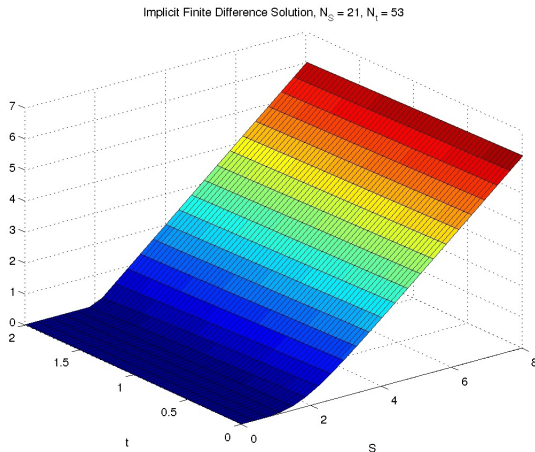
Note: the condition $\sigma^2 \geq r$ is sufficient but not necessary.

We compute the LU-factorization of $P\mathbf{A}$ (P is the permutation matrix) in $O(N_S)$ operations, and then use the resulting triangular factors to compute $\mathbf{V}^n$ at each time step in $O(N_S)$ operations. So the total cost is $O(N_t N_S)$ operations.

For the RHS vector $\mathbf{b}^n$, we determine V_0^n and $V_{N_S+1}^n$ using the boundary conditions, and we determine $\mathbf{V}^{N_t}$ from the value of the option at expiry.

Example on implicit FDM

Consider a call option with $r = 0.05$, $\sigma = 0.3$, $T = 2$, $S_{max} = 8$, and $K = 2$. Choose $N_S = 21$ and $N_t = 53$, and thus, $\Delta S = 8/22$ and $\Delta t = 2/53$.



Final remarks on backward-forward Euler methods

- The forward Euler scheme has overall accuracy $O(\Delta t + (\Delta S)^2)$ ($\max_{n,j} |V_j^n - V(S_j, t_n)| \leq C(\Delta t + (\Delta S)^2)$), which is also true for the backward Euler scheme.
- **Comparison:** In forward Euler method, a linear system must be solved at each time step—this is a disadvantage in terms of computational cost. However, the implicit scheme is unconditionally stable, with no restriction on the mesh ratio $\Delta t/(\Delta S)^2$.
- If σ (or r) is time-dependent, then the coefficient matrix $\mathbf{A}$ in the forward Euler method changes at each time step t_n . This increases the computational complexity when solving for the approximate solution $\mathbf{V}^n$ for each n .
- If $\sigma^2 \ll r$, the Black–Scholes PDE becomes convection-dominated. In such cases, the implicit scheme may produce nonphysical oscillations in the numerical solution.

Crank-Nicolson (CN) Method

The *CN method* is another type of *implicit* FDM, which is $O((\Delta t)^2 + (\Delta S)^2)$ accurate (offering higher accuracy in time compared to the previous two methods). It is based on approximating the PDE at the midpoint $(S_j, t_{n+\frac{1}{2}})$, i.e., at the half-time level between t_n and t_{n+1} ,

$$t_{n+\frac{1}{2}} = \frac{1}{2}(t_n + t_{n+1}) = t_n + \frac{1}{2} \Delta t.$$

From the Black-Scholes PDE:

$$\frac{\partial V}{\partial t}(S_j, t_{n+\frac{1}{2}}) + rS_j \frac{\partial V}{\partial S}(S_j, t_{n+\frac{1}{2}}) + \frac{1}{2}\sigma^2 S_j^2 \frac{\partial^2 V}{\partial S^2}(S_j, t_{n+\frac{1}{2}}) = rV(S_j, t_{n+\frac{1}{2}})$$

We use central difference for $\frac{\partial V}{\partial t}$ (with $\Delta t/2$ shift, $O((\Delta t)^2)$ accurate) and $\frac{\partial V}{\partial S}$ ($O((\Delta S)^2)$ accurate), and second central difference for $\frac{\partial^2 V}{\partial S^2}$ ($O((\Delta S)^2)$ accurate):

$$\begin{aligned} & \frac{V(S_j, t_{n+1}) - V(S_j, t_n)}{\Delta t} + rS_j \frac{V(S_{j+1}, t_{n+\frac{1}{2}}) - V(S_{j-1}, t_{n+\frac{1}{2}})}{2\Delta S} \\ & + \frac{1}{2}\sigma^2 S_j^2 \frac{V(S_{j+1}, t_{n+\frac{1}{2}}) - 2V(S_j, t_{n+\frac{1}{2}}) + V(S_{j-1}, t_{n+\frac{1}{2}})}{(\Delta S)^2} \approx rV(S_j, t_{n+\frac{1}{2}}), \end{aligned}$$

How should we handle $t_{n+\frac{1}{2}}$?

CN Method (continued)

From Taylor series expansion,

$$V(S_j, t_{n+1}) = V(S_j, t_{n+\frac{1}{2}}) + \frac{\Delta t}{2} \frac{\partial V}{\partial t}(S_j, t_{n+\frac{1}{2}}) + O(\Delta t^2)$$

$$V(S_j, t_n) = V(S_j, t_{n+\frac{1}{2}}) - \frac{\Delta t}{2} \frac{\partial V}{\partial t}(S_j, t_{n+\frac{1}{2}}) + O(\Delta t^2)$$

Adding these up, we observe that

$$V(S_j, t_{n+\frac{1}{2}}) \approx \frac{1}{2} (V(S_j, t_n) + V(S_j, t_{n+1})) \quad \text{with } O((\Delta t)^2) \text{ error}$$

(known as a **local average approximation**). So, the CN scheme is defined as:

$$\frac{V_j^{n+1} - V_j^n}{\Delta t} + r S_j \frac{V_{j+1}^{n+\frac{1}{2}} - V_{j-1}^{n+\frac{1}{2}}}{2\Delta S} + \frac{1}{2} \sigma^2 S_j^2 \frac{V_{j+1}^{n+\frac{1}{2}} - 2V_j^{n+\frac{1}{2}} + V_{j-1}^{n+\frac{1}{2}}}{(\Delta S)^2} = r V_j^{n+\frac{1}{2}},$$

for $j = 1, 2, \dots, N_S$ and $n = N_t - 1, \dots, 2, 1, 0$, where $V_j^{n+\frac{1}{2}} = \frac{1}{2}(V_j^n + V_j^{n+1})$.
The above CN method is $O((\Delta t)^2 + (\Delta S)^2)$ accurate.

Risk Management

Portfolios—comprising weighted combinations of calls and puts with varying parameters (e.g., strike prices and expiries)—are constructed to achieve specific financial objectives.

Delta Hedging: A common risk management strategy aiming to make the portfolio insensitive (i.e., zero first derivative) to small changes in the underlying asset price.

To implement delta hedging and other risk management strategies, we need to quantify how sensitive option values are to different market variables.

The Greeks: In options trading and risk management, the *Greeks* are key metrics used to measure the sensitivity of an option's price to changes in:

- S — the underlying asset price
- t — time
- σ — volatility
- r — the risk-free interest rate

Understanding and using the Greeks is essential for effectively assessing and managing the risks associated with options portfolios.

Risk Management (cont.), V is the value of the option

The main **Greeks** used in options risk management are:

- **Delta** ($= \frac{\partial V}{\partial S}$): Measures the rate of change of V w.r.t. the underlying asset price S . It reflects the sensitivity of the option price to small changes in S .
- **Gamma** ($= \frac{\partial^2 V}{\partial S^2}$): Measures the rate of change of Delta w.r.t. S . It indicates how Delta changes as the underlying price changes, and is crucial for assessing non-linear price risk.
- **Theta** ($= \frac{\partial V}{\partial t}$): Represents the sensitivity of V to the passage of time, commonly known as *time decay*.
- **Rho** ($= \frac{\partial V}{\partial r}$): Measures the sensitivity of V to changes in the risk-free interest rate r . Rho is particularly relevant for options on interest-rate-sensitive instruments such as bonds.
- **Vega** ($= \frac{\partial V}{\partial \sigma}$): Measures the sensitivity of V to changes in the volatility σ of the underlying asset.

Values of Delta, Gamma, and Theta can be computed using the Black–Scholes formula or estimated numerically using finite difference methods applied to the Black–Scholes PDE.